Answer Script Evaluation using OCR, Semantic Analysis and Generative AI

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Abstract—Marking student's descriptive answers is time consuming and can be labour intensive when answer scripts are hand written or only available as scanned documents. Automated grading has traditionally had difficulty in finding semantically correct answers using alternative wording. Recent advances in Optical Character Recognition, transformer language models, Large Language Models, semantic similarity and Retrieval-Augmented Generation provide opportunities to develop automated answer evaluation. This survey summarises the main findings on short answer grading, document text extraction, semantic representations, Large Language Model evaluation, retrieval based grounding and automatic feedback generation. Traditional methods relied on lexical similarity and manually designed features, while transformer-based methods improved contextual representation. There has been recent work on the possibility of using Large Language Models to evaluate open-ended answers, but concerns regarding consistency, explainability, reference grounding, and oversight by humans are still major. The use of Optical Character Recognition adds another layer of complexity, as errors in the recognition of hand-written or scanned scripts can have a direct impact on the evaluation that follows. Based on the survey, this paper identifies a research gap in end-to-end systems integrating answer-script processing, semantic evaluation, reference-aware retrieval, verification and feedback generation into a single workflow. This project tries to fill the research gap through three major modules, Answer Script Processing and OCR, Intelligent Answer Evaluation and Scoring, Feedback, Verification and Result Generation. It offers a structured and explainable AI-assisted academic assessment process, where human evaluation is embedded as a necessary validation layer.

Index Terms—Automatic Answer Evaluation, Automatic Short Answer Scoring, Optical Character Recognition, Semantic Similarity, Large Language Models, Retrieval-Augmented Generation, Educational Technology, AI-based Evaluation

I. INTRODUCTION

Assessment is an important part of the educational process, as it helps an instructor to evaluate students learning and gives an overview of what the students' performance was. Descriptive and short answer questions are a useful assessment tool for testing the understanding of concepts. They give students the chance to describe what they have learnt in their own words. Manual grading of written answers in a big class can be a time-consuming and cost-intensive task. In addition, human graders may have different interpretations of students' answers to questions in terms of paraphrasing, lack of explanation and description of the concepts.

Traditionally, automated grading systems have tackled the problem of matching paraphrased or conceptually similar responses with the help of keyword matching, statistical attributes, and lexical similarity. These metrics tend to be easy to implement, but can fail when a correct answer is given in a language that is very different from the language used for standard or benchmark answers.

With the emergence of transformer-based language models, NLP tasks have achieved a more profound semantic understanding. BERT showed contextual representations of sentences [1] and Sentence-BERT showed sentence-level similarities [2] for semantic similarity and comparison tasks. These models have enabled the development of automated short-answer graders that compare answers based on semantic similarity.

Recently, Large Language Models have been explored for the purpose of evaluating open-ended responses. Studies such as [4], [5] and [7] have explored generative large language models that evaluate student responses in ways that go beyond simple keyword analysis. However, this class of model can suffer from inconsistencies, prompt sensitivity, hallucinations and the need for appropriate source data.

There is also the problem of digitizing handwritten content when seeking to automatically assess answers that were originally written down in examination papers. In such cases, Optical Character Recognition needs to be performed so that it can convert the text into a machine readable form for evaluation. Modern OCR systems, such as PaddleOCR, provide end-to-end text detection, recognition, and document processing solutions [10]. But the quality of this recognition may depend on the style of handwriting, image distortion, page layout and other factors.

One of the main ways to reduce hallucination and improve contextual accuracy of language models is retrieval augmented generation [3]. The idea is to feed the language model with relevant information retrieved from an external

source. This can be used to feed useful information to improve the accuracy of answer evaluation models. Retrieval-based methods can help the language model to evaluate the conceptual correctness of student responses and give richer feedback.

The literature therefore points to several critical research areas studied in isolation: document scanning and OCR, semantic similarity, evaluation using large language models, and retrieval based generation of feedback. However very few works try to cover the whole workflow from the processing of a scanned document to the evaluation of the response and the giving of feedback.

The survey therefore focuses on these technologies and their practical implementations to be faced with evaluation. Based on the constraints from literature, the project framework is divided into three major modules, which are (1) Answer Script Processing and OCR, (2) Intelligent Answer Evaluation and Scoring, and (3) Feedback, Verification and Result Generation.

II. RELATED WORK

A. Conventional Automatic Short Answer Grading

The early Automatic Short Answer Grading (ASAG) systems were mainly based on comparing the student's answer with one or more reference answers. Lexical similarity, keyword overlap, TF-IDF representations and hand-crafted linguistic features were generally used for this comparison. The approach offers an interpretable basis for grading, but its capacity for correct grading is constrained by the strong reliance on the surface form of the answer when finding paraphrases.

For example, if a student explains the concepts correctly but in a substantially different word ordering or vocabulary than the reference, an exact match or a lexical similarity-based method would not recognize the correct explanation.

Semantic similarity methods try to measure how close are two responses in their meaning. Such methods have demonstrated that the embedding-based representations can provide a more flexible comparison between student answers and reference answers.

Automated short-answer grading has been achieved by frameworks like GradeAid, which blend lexical and semantic information. These approaches show that the combination of different types of evidence yields a more robust grading process than keyword matching alone. However, semantic similarity does not mean that an answer is entirely correct. Two sentences can be semantically related but differ in factual correctness or completeness. Therefore, besides similarity, answer evaluation should consider the question, the expected concepts and the grading criteria.

B. Semantic Evaluation Based on Transformer

The introduction of BERT has revolutionized natural language processing by providing contextual word embeddings using bidirectional transformer architectures [1]. BERT can represent the meaning of a word based on its surrounding context, unlike traditional word representations.

Later, Sentence-BERT was proposed for efficient sentence-level embeddings [2]. The semantic similarity computation between sentences makes it suitable for applications like duplicate detection, information retrieval, and answer comparison. Semantic embeddings can be used to determine whether a student's answer is conceptually similar to an expected answer, given the context of grading answers.

However, semantic similarity is not necessarily an indicator of completely correct answers. Two sentences can be semantically similar but different in terms of factual correctness or completeness. Thus, the evaluation of answers requires consideration of the question, concepts to be considered, and grade criteria, apart from similarity.

C. Assessing Responses Using LLMs

Large Language Models have brought a more flexible way of automated assessment. One could also prompt an LLM to evaluate the relationship between a question, reference answer, and student's answer, in addition to using metrics of numerical similarity.

In a study, Kortemeyer investigated the application of GPT-4 for automated grading of short answers and demonstrated that the model could reach significant agreement with human grading in the given context [4]. In an educational setting, Henkel et al. investigated the performance of LLMs in grading short answer questions [6]. These results suggest that LLMs may utilise more context information in evaluation than traditional methods.

Grévisse et al. investigated the application of large language models (LLMs) for grading in the setting of undergraduate medical education [5]. Their work suggests both the promise and limits of LLM-based assessment. LLMs results are a testament to their versatility, but highlight the essential function of human evaluation.

Pecuchova et al. performed a comparative study of generative AI models for open-ended questions in higher-educational environments [7]. Their work underscores the need to compare model outputs to human evaluations, rather than blindly accepting a given score.

Both studies emphasise the potential of LLMs as evaluation assistants, but also stress the importance of constraining their output using reference information, grading criteria and verification mechanisms.

D. Processing OCR Answer Scripts

The first challenge is to extract text from the answer scripts if they are in the form of scanned images, and Optical

Character Recognition converts the visual characters into machine-readable text and thus serves as an important pre-processing step for automated evaluation.

The PaddleOCR library offers a document oriented OCR framework with the capabilities of text detection, text recognition and document processing [10]. The pipeline can be applied on scanned academic documents before feeding the extracted text into downstream natural language processing components.

The quality of the grading depends directly on the quality of the OCR. If important words or sentences are misrecognized then the downstream semantic evaluation steps will be working on incomplete or incorrect information. Preprocessing, text normalization and appropriate handling of OCR uncertainty are therefore important components of an automated answer evaluation pipeline.

The OCR step is an additional headache for handwritten answer scripts too. Text extraction is more challenging than with conventional OCR for printed documents, due to variations in handwriting style, page quality, writing orientation, page background noise and overlapping marks.

E. Retrieval-Augmented Generation

Retrieval-Augmented Generation [3] is a method that combines information retrieval and a generative language model. Instead of a language model only, we can allow the model to access some external knowledge collection to fetch relevant information and provide these fetched details as context.

This method is applicable to education answering evaluation. Grading normally requires some domain knowledge. We can save the reference answers, lecture notes, marking rubrics or concept descriptions in a searchable knowledge base. While evaluating an answer, we can retrieve the relevant information and feed it to the evaluation model.

Vector databases can facilitate this approach by storing embeddings and retrieving semantically related information. The proposed framework involves storing reference embeddings in Qdrant and retrieving relevant content by means of semantic similarity.

Recent work has also explored graph-based retrieval for automated short-answer grading and feedback. GraphRAG-based approaches try to associate grading decisions with structured concepts and instructor-defined knowledge [8].

F. Generation of Automated Feedback

Assessment is not the whole of education. Learners gain advantages in identifying what they understood, what they missed in their answers and how to improve their answers.

Large language models can provide feedback by examining how a student's answer compares to the expected answer. However, without restriction, the feedback generated may not be sufficiently grounded in the subject matter. The generated feedback can be supported with evidence by using retrieval-based context.

Recent work has looked at interactive feedback in conjunction with automated short-answer grading [9]. These approaches underscore the value of coupling scoring with explanatory feedback instead of just showing a solitary automated grade.

III. COMPARATIVE ANALYSIS

The major approaches reviewed in the literature are summarized in Tables I and II. The comparison focuses on the main contribution of each method and the limitations that are relevant to the automated answer evaluation task.

TABLE I

COMPARISON OF EXISTING APPROACHES FOR AUTOMATED ANSWER EVALUATION

No.	**Author & Year**	**Inference**	**Disadvantages**
1	Devlin et al., 2019	Bidirectional transformer representations were introduced for enhanced contextual understanding in the field of NLP.	Not explicitly developed for answer grading and requires specific customization for the task
2	Reimers and Gurevych, 2019	An efficient model for generating sentence embeddings for semantic similarity and comparison tasks was suggested.	The semantic similarity between sentences does not guarantee the factual correctness and completeness of the information
3	GradeAid	A system combining text features for automatic short-answer scoring was suggested.	The results strongly depend on the used training data and reference information
4	Kortemeyer, 2024	A method based on GPT-4 for automated scoring of student responses was investigated.	The results significantly depend on the prompts and the used grading instructions
5	Grévisse et al., 2024	LLMs for automatic grading in undergraduate medical education were investigated.	The results showed that human agreement was not perfect, and human oversight is still required.

TABLE II

COMPARISON OF EXISTING APPROACHES FOR AUTOMATED ANSWER EVALUATION (CONTINUED)

No.	Author & Year	Inference	Disadvantages
6	Henkel et al., 2024	Explored the application of large language models (LLMs) to grade short-answer questions in K-12 education.	Limited generalizability and consistency in different educational environments.
7	Pecuchova et al., 2025	Compared generative AI models and semantic approaches for open ended questions.	Varied model performance across questions and evaluation criteria.
8	Lewis et al., 2020	Integrates retrieval and generation for external knowledge context.	Requires appropriate knowledge sources and effective retrieval mechanisms.
9	Xu, Corter, 2026	Grounded grading and feedback with structured knowledge and retrieval.	Needs well constructed knowledge structures and teaching materials.
10	AI-Based Automated Grading, 2026	Combines automated evaluation and feedback for short-answer responses.	Requires further validation for different answer types and educational contexts.

IV. PROPOSED SYSTEM

The review of literature indicates that automated answer assessment requires multiple similarity or language models. The system should be able to view the answer script, recognize the student's answer, comprehend its meaning in context of the appropriate reference material and provide meaningful feedback.

The proposed framework incorporates Optical Character Recognition, semantic analysis, vector-based retrieval, Large Language Models and verification into a unified workflow. The system consists of three main modules.

A. Module 1: Script Processing and OCR of Answers

The first module takes scanned or photographed answer scripts and converts them to structured, machine readable text. The input answer sheet is first preprocessed for document to improve the quality of input image. Then the text regions are identified and respective characters are recognized using OCR.

The main OCR framework used for text detection and recognition is PaddleOCR. The extracted content is normalized and structured along question and answer boundaries, which allows separate processing of individual answers.

B. Module 2: Intelligent Grading and Scoring of Answers

The second module assesses each extracted response by semantic representations, reference information, retrieval and generative AI. The student's response is judged in relation to the question and the reference information.

Semantic embeddings enable us to locate conceptually related information even when the student's choice of words does not match the reference answer. A vector database like Qdrant can be used to retrieve reference information.

That retrieved information is then supplied to the language model as context evidence. The evaluation is based on conceptual coverage, correctness, completeness and relevance and not merely keyword matching.

An orchestration component is responsible for the retrieval, semantic analysis, and language model evaluation stages.

C. Module 3: Verification, Results Production and Feedback

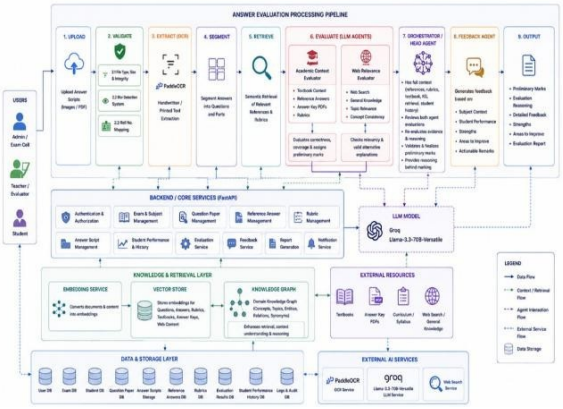


Fig 1. Overall Architecture of the Proposed System

V. IMPLEMENTATION

The proposed framework is implemented as a modular processing pipeline where each stage performs a specific function in the answer evaluation process. It implements the conversion of answer scripts into machine readable content, the evaluation of the extracted responses using semantic and reference-based techniques, and the generation of structured feedback.

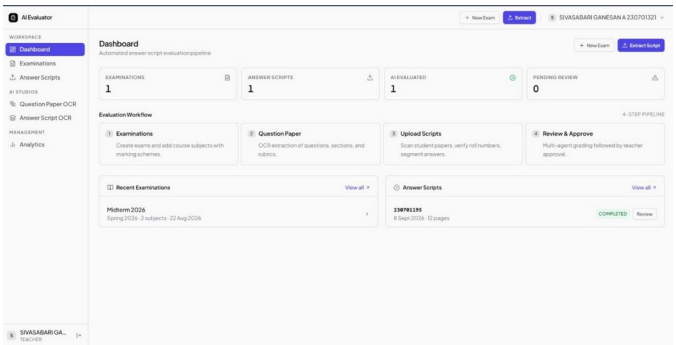


Fig 2. Main Dashboard for Evaluating Answer Scripts of Proposed System

A. Module 1: OCR and Processing of Answer Scripts

The implementation starts with the uploaded answer script, which can be uploaded in the form of scanned document or image. The input document is first pre-processed to improve its quality before text recognition.

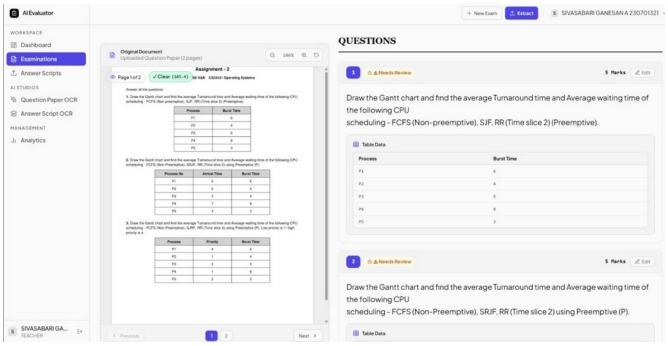


Fig 3. Question Paper Extraction Interface

PaddleOCR is used to detect and recognize text. The OCR pipeline finds regions of text and translates the detected content into machine-readable text. The extracted text is then normalized to reduce the variations and errors introduced during the process of OCR.

The processed text is structured by the questions and answers that correspond to it. This separation enables the evaluation pipeline to process each student response individually.

Therefore, OCR is integrated directly into the evaluation pipeline rather than treated as a separate preprocessing operation. Recognition errors may affect subsequent semantic analysis, so the quality of extracted text is an important part of the overall evaluation process.

B. Module 2: Smart Evaluation and Grading of Answers

The next step is to assess the extracted student answers in relation to the corresponding question and reference information.

The student's response and relevant reference material are used to generate semantic embeddings. Cosine similarity can be used to measure the semantic similarity between student response and a reference representation. The third module produces the final evaluation output. The evaluation result contains the score given, an explanation, feedback and information about confidence.

The feedback part helps to know the strengths, missing concepts, and improvement areas. Retrieved reference information can be given during feedback generation to be consistent with the expected content.

There is a verification stage to detect evaluations that may need further validation. NVIDIA NIM can be added as an additional model-serving and verification component wherever required by the evaluation workflow.

The final output is structured so that both students and evaluators can understand the grading decision and the corresponding feedback.

The semantic similarity between the two representations can be calculated using cosine similarity:

$$S_{\text{semantic}} = \frac{E_s \cdot E_r}{\|E_s\| \, \|E_r\|} \quad (1)$$

where Es is the embedding of the student response and Er is the embedding of the relevant reference information.

The calculated similarity is taken as one piece of evidence, not the only grading criterion. Conceptual coverage, correctness and completeness and the requirements of the question are also part of the evaluation process.

Referring information is kept in vector representations in Qdrant. For each student response, we can fetch semantically relevant reference content from the vector database. The retrieved content is then provided as context to the language model to take into account.

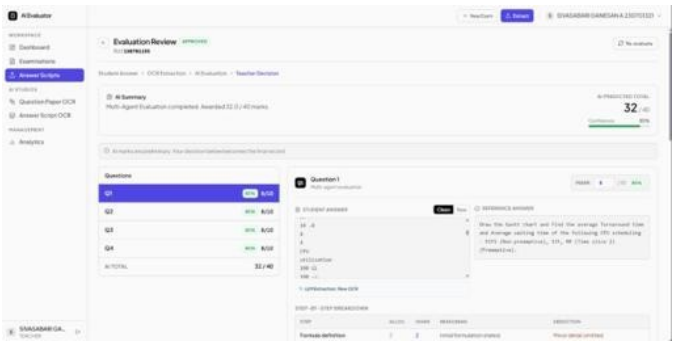


Fig 4. AI for Mark Allocation and Answer Evaluation

The orchestration component coordinates the retrieval, semantic-analysis and language-model stages. Instead of a single similarity score or an isolated language model response this provides a structured workflow for evaluation.

Thus, the evaluation stage focuses on whether the expected concepts are meaningfully represented in the student's answer, not on requiring the student to reproduce the exact wording of the reference answer.

C. Module 3: Feedback, Verification and Results Generation

The third module gives the final result for the student or evaluator after the initial evaluation.

The feedback part analyzes the evaluation result to identify concepts that were correctly addressed, information that is missing, and areas needing improvement. The system provides not only a numerical score but also an explanation of the basis of the evaluation.

The retrieved reference information can be included in the feedback generation so that the explanation is still connected to the expected concepts and course related information.

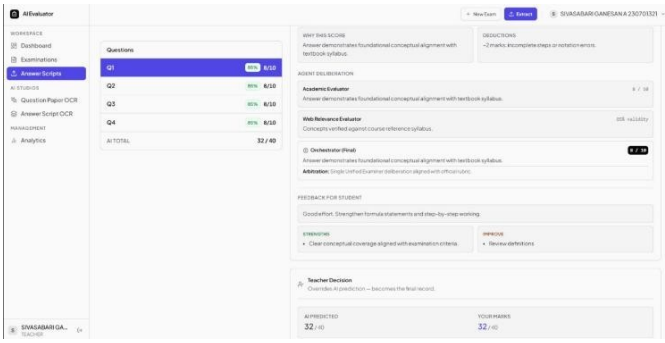


Fig 5. Feedback to Students Generated by AI

It includes a verification step to mark evaluations that may require further validation. When activated in the evaluation workflow, NVIDIA NIM could be used as a supplementary verification or a model-serving layer. The goal of this stage is not to equally weigh the reliability of each created evaluation.

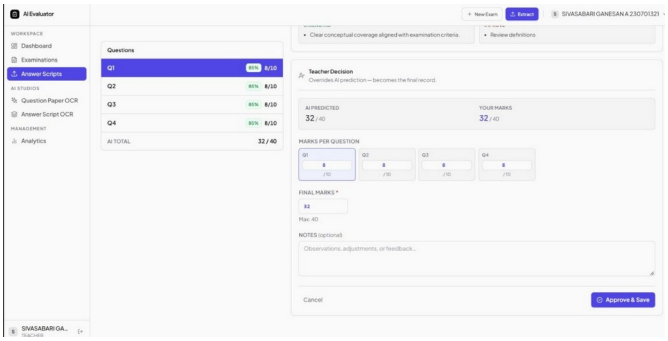


Fig 6. Interface for Teacher Review and Final Mark Approval.

The final output consists of the assigned score, evaluation explanation, feedback and confidence related information. The structured output helps the student to understand the evaluation and the concepts that need further improvement.

VI. CONCLUSION

The automatic evaluation of descriptive student answers has moved from simple keyword matching to semantic, transformer-based and Large Language Model approaches. The surveyed literature shows that semantic representations are useful for paraphrased answers than keyword matching and LLM-based methods are more flexible to evaluate open-ended answers. OCR is still needed when the answer script input is not available as digital text, while retrieval based methods provide a mechanism to ground model decisions on relevant reference information.

The survey discusses some shortcomings of current approaches. OCR errors can propagate to the final evaluation, semantic similarity is not enough to decide correctness, and LLM based evaluation depends on appropriate prompting, reference information and verification. Furthermore, good assessment doesn't just consist of a number, but also involves feedback.

According to the survey, the AI Answer Evaluator is divided into three major modules: Answer Script Processing and OCR, Intelligent Answer Evaluation and Scoring, Feedback, Verification and Result Generation. The proposed approach combines OCR, semantic similarity, vector retrieval, generative AI and verification in a single evaluation workflow.

The system is intended to support the grading process and not to replace human graders. Human evaluation continues to be important for ambiguous and diagrammatic answers and for high stakes academic evaluation. Future work includes improving handwriting recognition, support for answers based on diagrams and graphs, multi lingual evaluation, calibration of grading rubrics and confidence aware human validation.

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